

Retail Sales Data Analysis Using PySpark

Duration: 1 Hour

Level: Medium

Individual Assessment

1. Business scenario

You are working as a Junior Data Engineer at a retail company that operates multiple stores. The company collects daily sales transaction data from its stores and wants to analyze sales performance, identify popular products, and understand customer purchasing patterns.

The sales data is stored in a CSV file. Your task is to use PySpark DataFrames to process the data, perform aggregations, and generate meaningful business insights.

2. Dataset

Create a CSV file named `retail_sales.csv` with the following data.

Sample dataset — 15 transaction records

Transaction_ID	Customer	Product	Category	Store	Quantity	Unit_Price
T101	C001	Laptop	Electronics	Kochi	1	55000
T102	C002	Mouse	Accessories	Trivandrum	2	500
T103	C003	Keyboard	Accessories	Kochi	1	1500
T104	C001	Headphones	Electronics	Calicut	2	2000
T105	C004	Laptop	Electronics	Trivandrum	1	55000
T106	C005	Mouse	Accessories	Kochi	3	500

T107	C002	Monitor	Electronics	Calicut	1	12000
T108	C006	Keyboard	Accessories	Trivandrum	2	1500
T109	C003	Laptop	Electronics	Kochi	1	55000
T110	C004	Mouse	Accessories	Calicut	4	500
T111	C007	Monitor	Electronics	Kochi	2	12000
T112	C001	Keyboard	Accessories	Calicut	1	1500
T113	C005	Headphones	Electronics	Trivandrum	1	2000
T114	C006	Mouse	Accessories	Kochi	2	500
T115	C007	Laptop	Electronics	Calicut	1	55000

3. Tasks to be completed

Section A – Data loading and inspection (10 marks)

1. Load the CSV file into a PySpark DataFrame.
2. Display the first five records.
3. Display the schema of the DataFrame.
4. Find the total number of transactions in the dataset.

Section B – Data transformation (15 marks)

5. Create a new column named Total_Amount by multiplying Quantity by Unit_Price.
6. Display only the transactions where Total_Amount is greater than ₹10,000.
7. Identify and display any duplicate transaction records, if present.

Section C – Business analysis (20 marks)

8. Calculate the total revenue generated by each product.
9. Calculate the total revenue generated by each store.
10. Find the total quantity sold for each product category.

11. Identify the product that generated the highest total revenue.
12. Calculate the average transaction amount for each store.

Section D – Advanced analysis (5 marks)

13. Find the top three customers based on their total spending.

4. Expected deliverables

- A Python notebook (.ipynb) containing the PySpark solution.
- Output displayed for all 13 tasks.
- A short conclusion identifying the best-performing store and highest-revenue product.

5. Suggested time allocation

Activity	Time
Load and inspect data	10 minutes
Transform data	15 minutes
Business analysis	25 minutes
Advanced analysis and review	10 minutes
Total	60 minutes

Assessment rule: Use PySpark DataFrame operations for all data processing and analysis. Do not convert the complete dataset into Pandas for solving the questions.